Nonlinear functions — Part 1

Advanced Math · Math · 86 questions

1.1 b58ee74d-9a79-4ad3-b997-7fb216ab725f · Easy

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company's scheduled deliveries over 8 months, where y is the estimated number of scheduled deliveries x months after the end of May 2012, where $0 \leq x \leq 8$. Which statement is the best interpretation of the y-intercept of the graph of this equation in the xy-plane?

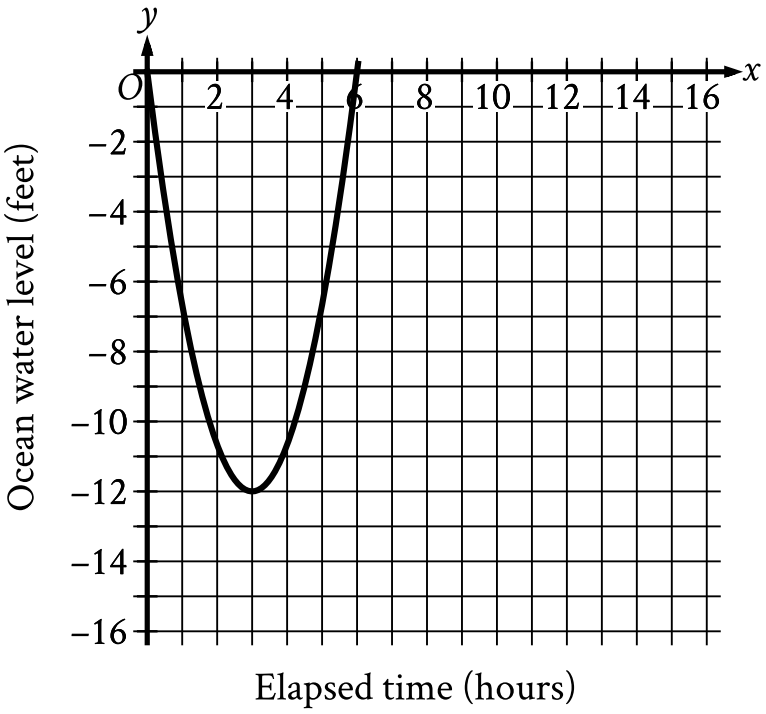
A. At the end of May 2012, the estimated number of scheduled deliveries was 0.

B. At the end of May 2012, the estimated number of scheduled deliveries was 29.

C. At the end of June 2012, the estimated number of scheduled deliveries was 0.

D. At the end of June 2012, the estimated number of scheduled deliveries was 29.

1.2 1549b7a3-5fbf-46f8-8861-103e5d9edae0 · Easy



- The parabola opens upward.
- The parabola passes through the following points:
  - (0 comma 0)
  - (3 comma negative 12)
  - (6 comma 0)

Scientists recorded data about the ocean water levels at a certain location over a period of 6 hours. The graph shown models the data, where $y = 0$ represents sea level. Which table gives values of x and their corresponding values of y based on the model?

A.

x	y
0	-12
0	3
3	6

B.

x	y
0	0
3	12
0	-6

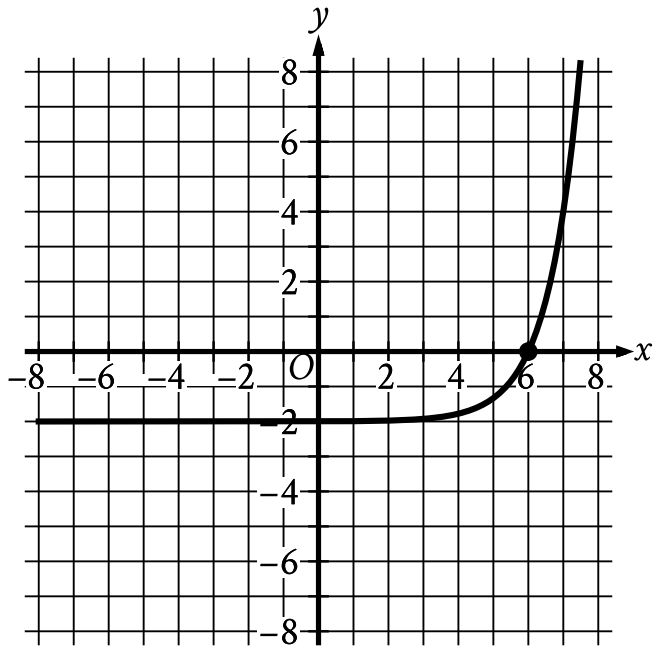
C.

x	y
0	0
3	-12
6	0

D.

x	y
0	0
12	3
-6	0

1.3 43c1b9bd-91a3-4f33-8ec6-ca10f3f10196 · Easy

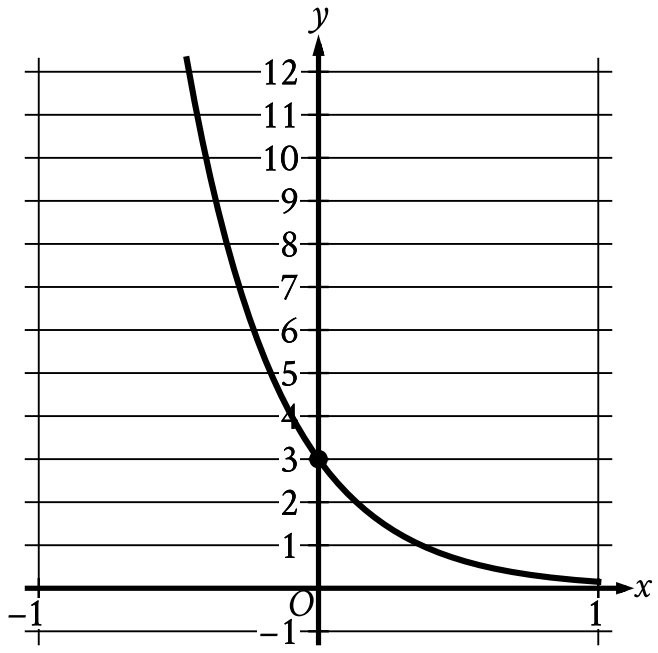


- Moving from left to right:
  - The curve passes from quadrant 3 to quadrant 4 to quadrant 1.
  - In quadrant 3, the curve trends up gradually to the y axis.
  - In quadrant 4, the curve trends up gradually to the point (6 comma 0).
  - In quadrant 1, the curve trends up sharply.
- As x decreases, the curve approaches the line y equals negative 2.
- The curve passes through the following points:
  - (5 comma negative four thirds)
  - (6 comma 0)

What is the *x*-coordinate of the *x*-intercept of the graph shown?

Student-produced response: _____

1.4 fc448e0e-6144-425a-a9b3-2ff8ee674f1b · Easy



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends down sharply to the point (0 comma 3).
  - In quadrant 1, the curve trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (0 comma 3)
  - approximately (0.5 comma 0.7)

The graph of the exponential function f is shown, where $y = fx$. The y-intercept of the graph is $0, y$. What is the value of y?

Student-produced response: _____

1.5 e0bc2b05-f917-4d64-b9ad-36fa20c0b430 · Easy

The function f is defined by $fx = 8\sqrt{x}$. For what value of x does $fx = 48$?

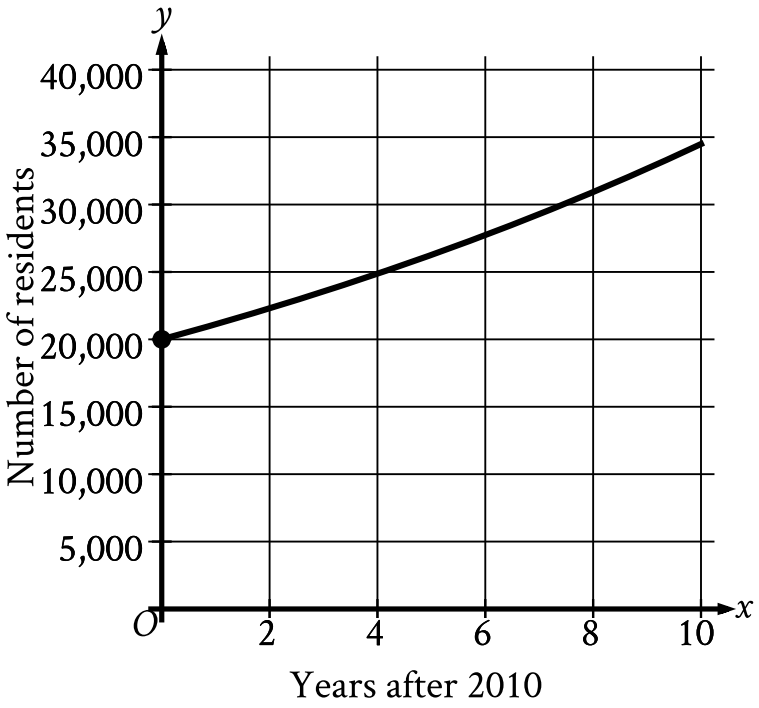
A. 6

B. 8

C. 36

D. 64

1.6 58bfb590-016f-4bec-8d23-459d22e7482a · Easy



- The curve is in quadrant 1.
- The curve trends gradually up from left to right.
- The curve begins at the point (0 comma 20,000).
- The curve passes through the following points:
  - (0 comma 20,000)
  - approximately (5 comma 26,263)
  - approximately (10 comma 34,488)

The graph shown models the number of residents of a certain city x years after 2010. How many residents does this model estimate the city had in 2010?

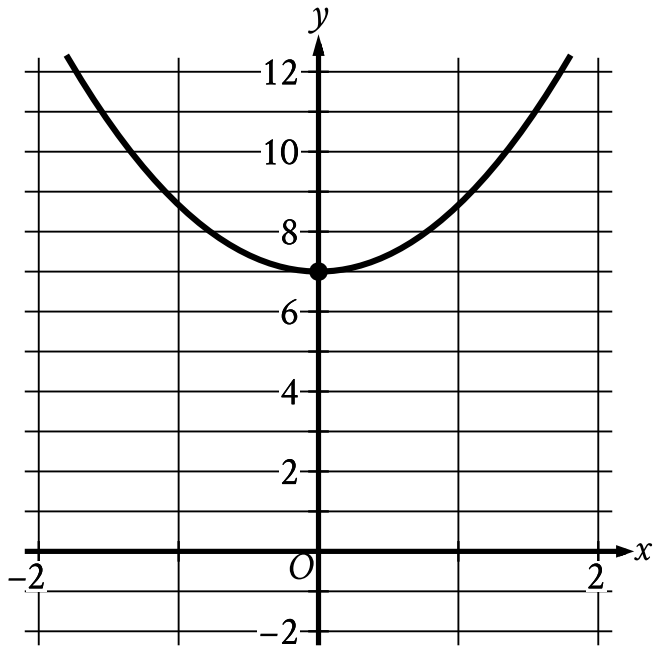
A. 0

B. 2,000

C. 20,000

D. 25,000

1.7 752821ca-8045-45d2-afe6-e57b8760f5f1 · Easy



- The parabola opens upward.
- The vertex is at the point (0 comma 7).
- The parabola passes through the following points:
  - (negative 1 comma StartFraction 26 Over 3 EndFraction)
  - (0 comma 7)
  - (1 comma StartFraction 26 Over 3 EndFraction)

The parabola shown intersects the y-axis at the point x, y. What is the value of y?

Student-produced response: _____

1.8 84e5072e-3d6b-4453-a0ee-23b7b7999691 · Easy

The function k is defined by $kx = x^3 + 110$. What is the value of kx when $x = 2$?

A. 118

B. 116

C. 115

D. 110

1.9 e9d27a82-510c-4c48-a723-ee345529a8fb · Easy

$$Pt = 1{,}8001.02^t$$

The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. What is the best interpretation of $P0 = 1{,}800$ in this context?

A. The estimated number of marine mammals in the area was 102 when the study began.

B. The estimated number of marine mammals in the area was $1{,}800$ when the study began.

C. The estimated number of marine mammals in the area increased by 102 each year during the study.

D. The estimated number of marine mammals in the area increased by $1{,}800$ each year during the study.

1.10 3212a20d-5aee-41d8-8981-4e3ce65c7918 · Easy

The number of cells of a certain bacteria doubles every 25 minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of 830 cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

A. $fx = 830\frac{1}{2}^{\frac{x}{25}}$

B. $fx = 8302^{\frac{x}{25}}$

C. $fx = 830\frac{1}{3}^{\frac{x}{25}}$

D. $fx = 8303^{\frac{x}{25}}$

1.11 f827079b-37e0-4696-9873-897b36b24545 · Easy

The function g is defined by $gx = \sqrt{x} + 300$. What is the value of gx when $x = 81$?

A. 9

B. 300

C. 309

D. 381

1.12 cecc8739-2d4c-4044-8624-baee7ca16ce6 · Easy

The function g is defined by $gx = x^2 + 9$. For which value of x is $gx = 25$?

A. 4

B. 5

C. 9

D. 13

1.13 f45f8ef3-75e8-4989-991d-8bcf481614f7 · Easy

The kinetic energy, in joules, of an object with mass 9 kilograms traveling at a speed of v meters per second is given by the function K, where $Kv = \frac{9}{2}v^2$. Which of the following is the best interpretation of $K34 = 5{,}202$ in this context?

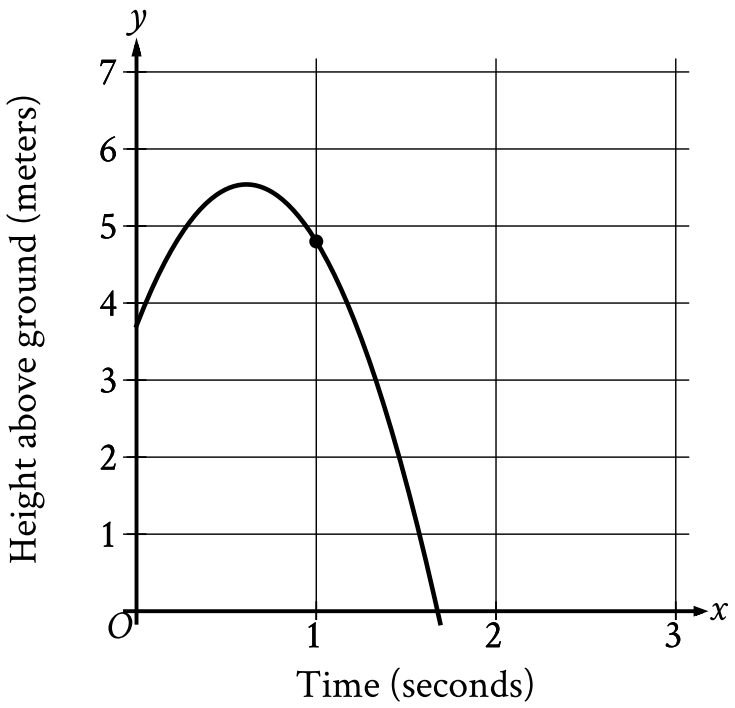
A. The object traveling at 34 meters per second has a kinetic energy of $5{,}202$ joules.

B. The object traveling at 340 meters per second has a kinetic energy of $5{,}202$ joules.

C. The object traveling at $5{,}202$ meters per second has a kinetic energy of 34 joules.

D. The object traveling at $23{,}409$ meters per second has a kinetic energy of 34 joules.

1.14 75b672c9-de51-448e-b5a4-f2bd63b15e93 · Easy



- All values are approximate.
- The parabola opens downward.
- The vertex of the parabola is at (0.6 comma 5.5).
- The parabola passes through the following points:
  - (0.0 comma 3.8)
  - (0.6 comma 5.5)
  - (1.0 comma 4.8)
  - (1.7 comma 0.0)

The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point $1.0, 4.8$ in this context?

A. 1.0 second after being launched, the ball's height above ground is 4.8 meters.

B. 4.8 seconds after being launched, the ball's height above ground is 1.0 meter.

C. The ball was launched from an initial height of 1.0 meter with an initial velocity of 4.8 meters per second.

D. The ball was launched from an initial height of 4.8 meters with an initial velocity of 1.0 meter per second.

1.15 29fc6f6a-c78f-4294-a4fc-98e4343406e7 · Easy

An investment account was opened with an initial value of $\$\,890$. The value of the account doubled every 10 years. Which equation represents the value of the account Mt, in dollars, t years after the account was opened?

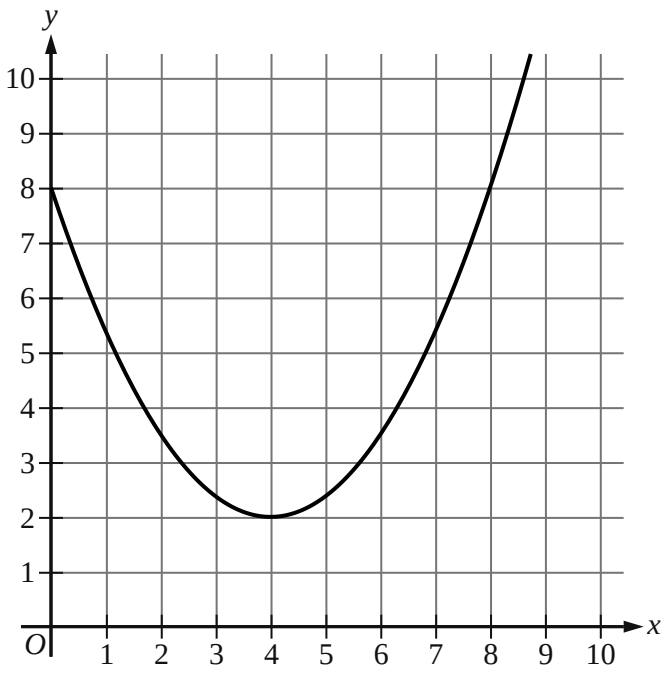
A. $Mt = 890\frac{1}{2}^{\frac{t}{10}}$

B. $Mt = 890\frac{1}{10}^{\frac{t}{2}}$

C. $Mt = 8902^{\frac{t}{10}}$

D. $Mt = 89010^{\frac{t}{2}}$

1.16 e07cc332-1b7d-4d38-a472-8db2ff2d4e41 · Easy



- The parabola opens upward.
- The vertex is at the point (4 comma 2).
- The parabola passes through the following points:
  - (0 comma 8)
  - (4 comma 2)
  - (8 comma 8)

The graph shows a marble's height above the ground y, in inches, x seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the y-intercept of the graph?

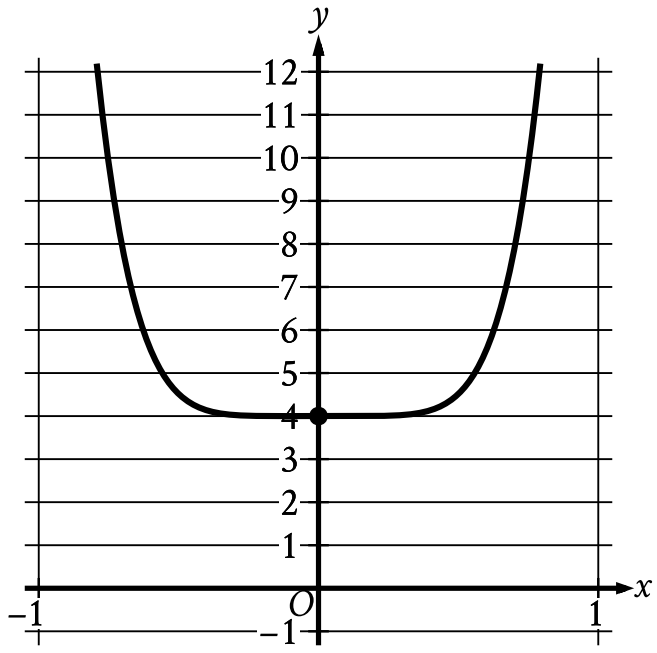
A. The marble's height was 0 inches above the ground 8 seconds after it started moving.

B. The marble's height was 8 inches above the ground when it started moving.

C. The marble's minimum height was 0 inches above the ground.

D. The marble's minimum height was 8 inches above the ground.

1.17 d6f230e9-a5fd-4506-837c-cde890c380bc · Easy

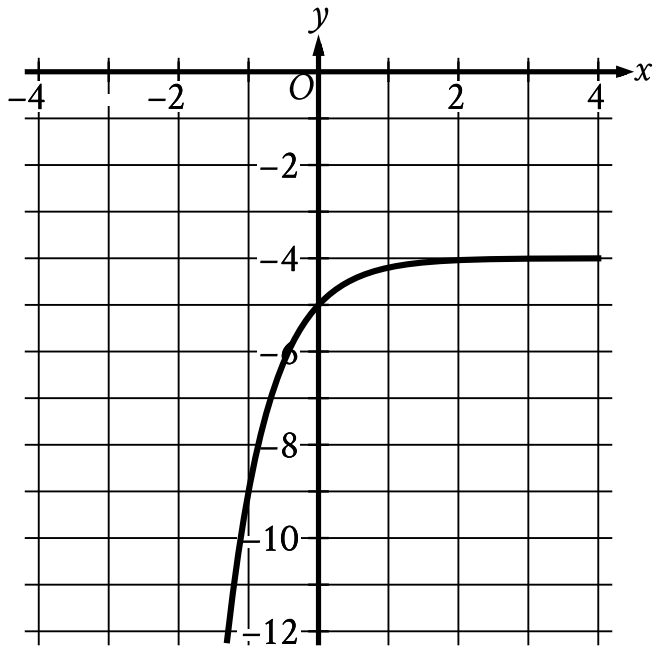


- In quadrant 2, the curve falls to its minimum at the point (0 comma 4).
- In quadrant 1, the curve then rises.

The graph of the polynomial function f, where $y = fx$, is shown. The y-intercept of the graph is $0, y$. What is the value of y?

Student-produced response: _____

1.18 37ced69c-527a-4d31-a90e-19a862272659 · Easy



- Moving from left to right:
  - The curve passes from quadrant 3 to quadrant 4.
  - In quadrant 3, the curve trends up sharply to point (0 comma negative 5).
  - In quadrant 4, the curve trends up gradually.
- As x increases, the curve approaches the line y equals negative 4.
- The curve passes through the following points:
  - (negative 1 comma negative 9)
  - (0 comma negative 5)
  - (1 comma negative 4.2)

What is the y-intercept of the graph shown?

A. $-1, -9$

B. $0, -5$

C. $0, -4$

D. $0, 0$

1.19 d6c4b6c3-23fb-477d-aba3-9ed20b31fbb9 · Easy

$$Pt = 24.81.036^{t}$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement "$P8$ is approximately equal to 32.91"?

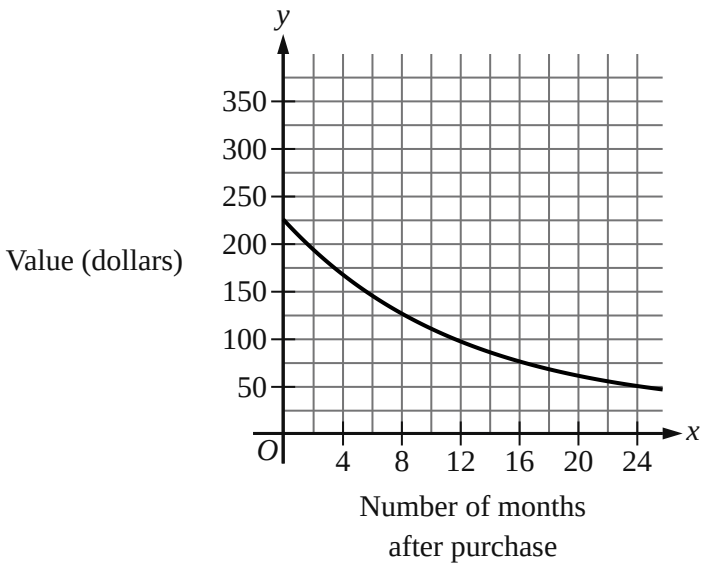
A. In 1984, the predicted population of this country was approximately 8 million.

B. In 1984, the predicted population of this country was approximately 32.91 million.

C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.

D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

1.20 d73b26c3-ce67-4987-8411-3f6ae5fa6ca3 · Easy



- Moving from left to right:
  - The curve is in quadrant 1.
  - The curve trends down gradually.
- The curve passes through the following approximate points:
  - (0 comma 225)
  - (24 comma 50)

The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

A. The estimated value of the tablet was $\$ 225$ when it was purchased.

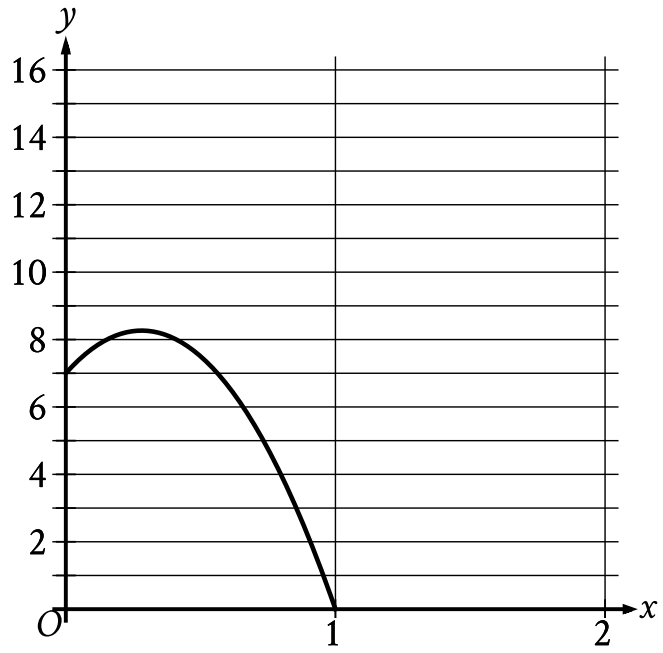
B. The estimated value of the tablet 24 months after it was purchased was $\$ 225$.

C. The estimated value of the tablet had decreased by $\$ 225$ in the 24 months after it was purchased.

D. The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

1.21 dc882084-e2b9-4e68-92b2-78a7a8f89acd · Easy

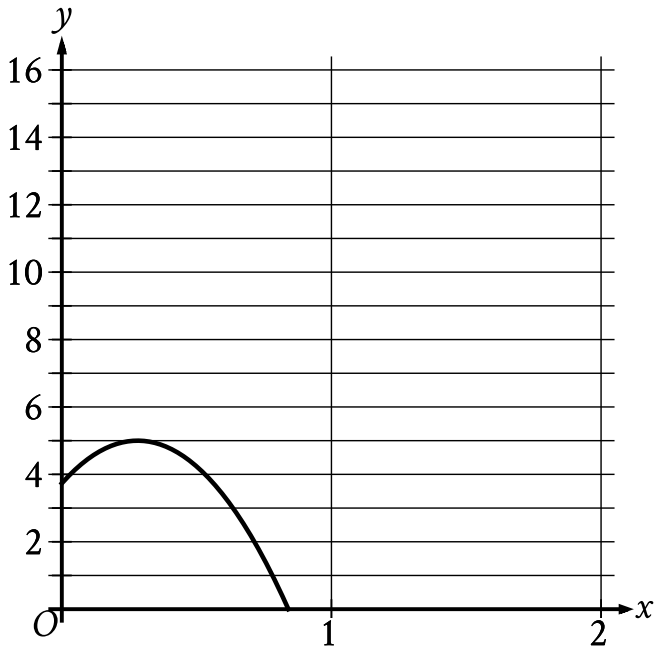
During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height y, in feet, of the ball x seconds after it was launched during the first part of the experiment.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 8.3).
- The parabola passes through the following points:
  - (0 comma 7)
  - approximately (0.3 comma 8.3)
  - approximately (0.6 comma 7)

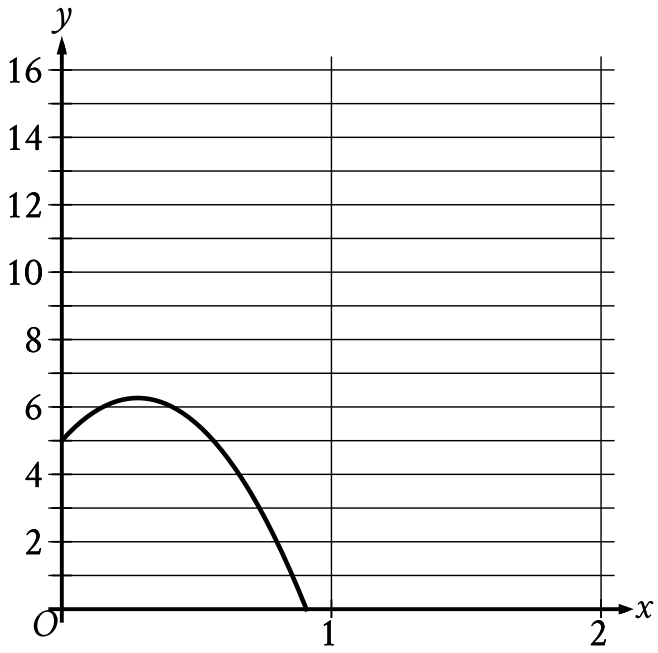
During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height y, in feet, of the ball x seconds after it was launched during the second part of the experiment?

A.



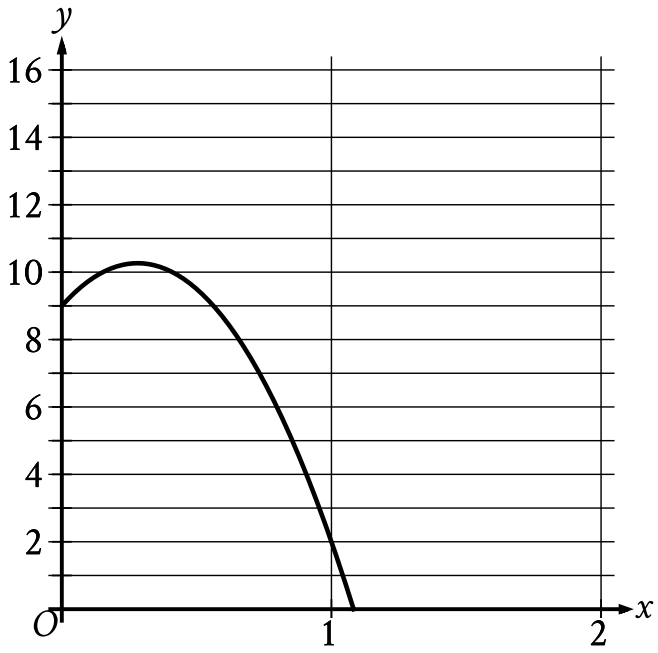
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 5.0).
- The parabola passes through the following points:
  - approximately (0 comma 3.7)
  - approximately (0.3 comma 5.0)
  - approximately (0.6 comma 3.7)

B.



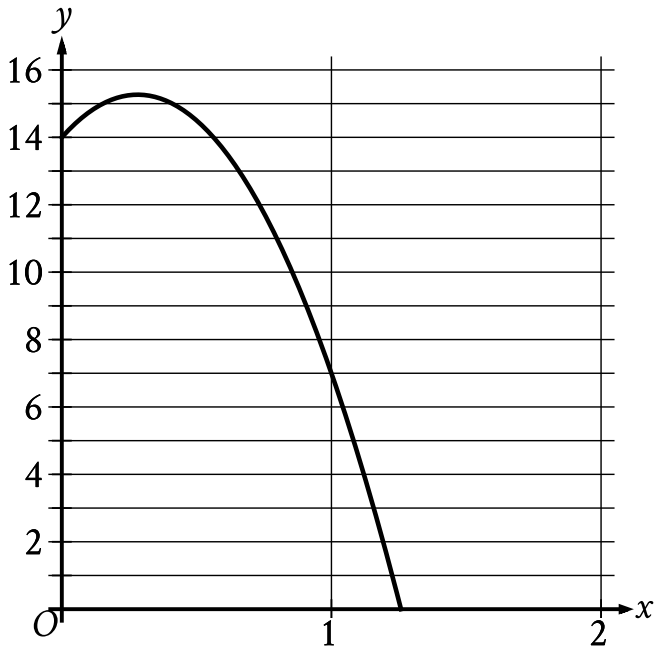
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 6.3).
- The parabola passes through the following points:
  - (0 comma 5)
  - approximately (0.3 comma 6.3)
  - approximately (0.6 comma 5)

c.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 10.3).
- The parabola passes through the following points:
  - (0 comma 9)
  - approximately (0.3 comma 10.3)
  - approximately (0.6 comma 9)

D.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 15.3).
- The parabola passes through the following points:
  - (0 comma 14)
  - approximately (0.3 comma 15.3)
  - approximately (0.6 comma 14)

1.22 e599f748-6055-4c3c-83f8-5366aad721b3 · Easy

A ball is dropped from an initial height of 22 feet and bounces off the ground repeatedly. The function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. Which equation defines h, where $h(n)$ is the estimated maximum height of the ball after it has hit the ground n times and n is a whole number greater than 1 and less than 10?

A. $h(n) = 22(0.22)^n$

B. $h(n) = 22(0.85)^n$

C. $h(n) = 850.22^n$

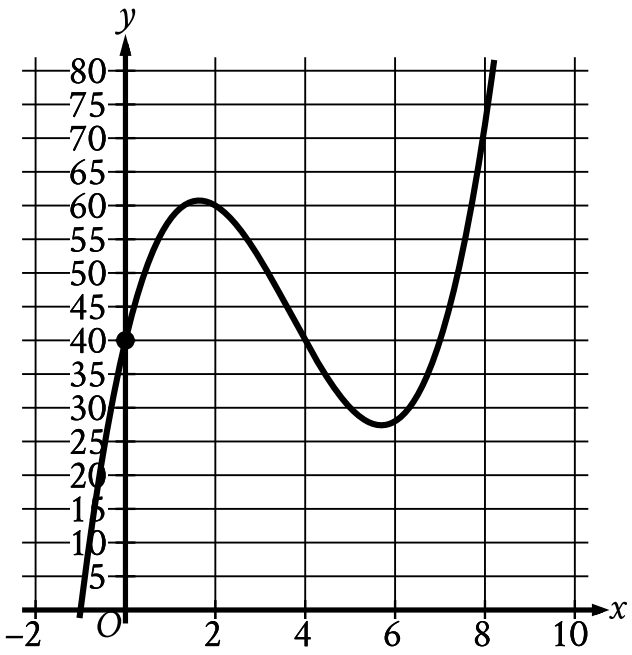
D. $h(n) = 85(0.85)^n$

1.23 a6fb9475-3880-4ae5-952c-4bbb83ee59af · Easy

The function f is defined by $f(x) = 6 + \sqrt{x}$. What is the value of $f36$?

Student-produced response: _____

1.24 591c966e-550b-4b07-97bb-f659afe623d8 · Easy



- Moving from left to right, the curve passes from quadrant 2 to quadrant 1.
- The curve has 1 relative maximum and 1 relative minimum.
- The curve passes through the following points:
  - (0 comma 40)
  - (4 comma 40)
  - (7 comma 40)

The y-intercept of the graph shown is x, y. What is the value of y?

Student-produced response: _____

1.25 4e837070-2e7d-4f56-993a-9d8d616e345b · Easy

$$hx = x^2 - 3$$

Which table gives three values of x and their corresponding values of hx for the given function h?

A.

x	1	2	3
hx	4	5	6

B.

x	1	2	3
hx	-2	1	6

C.

x	1	2	3
hx	-1	1	3

D.

x	1	2	3
hx	-2	1	3

1.26 73405bf8-4af6-4fad-9344-e38505f18416 · Easy

$$gx = 11\frac{1}{12}^{x}$$

If the given function g is graphed in the xy-plane, where $y = gx$, what is the y-intercept of the graph?

A. $0, 11$

B. $0, 132$

C. $0, 1$

D. $0, 12$

1.27 d38233c5-51bc-4d5b-b405-07bfe9387efc · Easy

The function $f(x) = 200{,}0001.21^{x}$ gives a company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online, where $0 < x \leq 10$. What is the best interpretation of the statement "$f5$ is approximately equal to $518{,}748$" in this context?

A. 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately $518{,}748$ dollars.

B. 5 years after the company started selling light bulbs online, its predicted annual revenue will have increased by a total of approximately $518{,}748$ dollars.

C. When the company's predicted annual revenue is approximately $518{,}748$ dollars, it is 5 times the predicted annual revenue for the previous year.

D. When the company's predicted annual revenue is approximately $518{,}748$ dollars, it is 5% greater than the predicted annual revenue for the previous year.

1.28 b0bf4cfd-4fbe-4424-acc0-d7a7f44468f7 · Easy

The function f is defined by $fx = x^{3} + 15$. What is the value of $f2$?

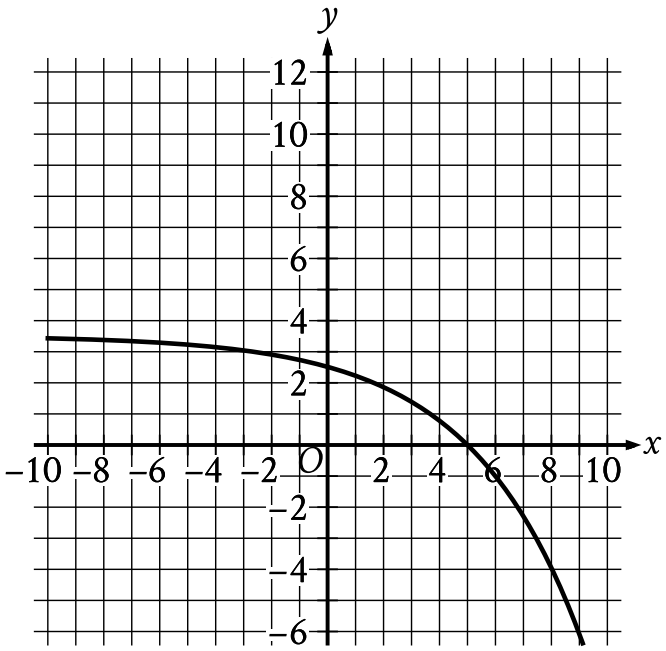
A. 20

B. 21

C. 23

D. 24

1.29 0eaf53b0-0c3b-45a2-bf0d-c423f1999a0d · Easy



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
  - In quadrant 2, the curve trends down gradually to the approximate point (0 comma 2.5).
  - In quadrant 1, the curve trends down sharply to point (5 comma 0).
  - In quadrant 4, the curve trends down sharply.
- The curve passes through the following points:
  - approximately (0 comma 2.5)
  - approximately (4 comma 0.8)

What is the *x*-intercept of the graph shown?

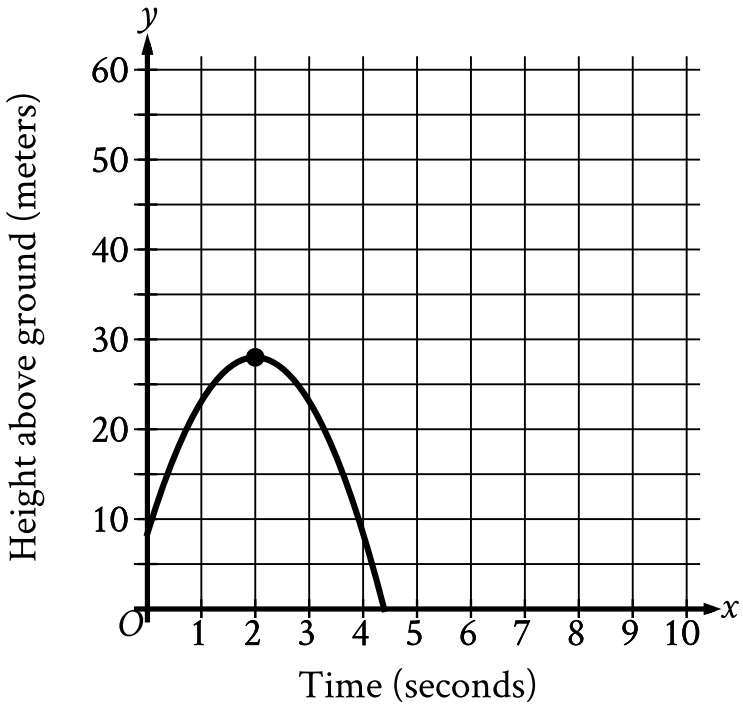
A. $-5, 0$

B. $5, 0$

C. $-2, 0$

D. $2, 0$

1.30 518e7cca-1f28-4828-8926-524f47cebb81 · Easy



- The parabola opens downward.
- The vertex is at point (2 comma 28).
- The parabola passes through the following points:
  - (0 comma 8.4)
  - (2 comma 28)
  - (4 comma 8.4)

An object was launched upward from a platform. The graph shown models the height above ground, y, in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

A. From $x = 0$ to $x = 2$

B. From $x = 0$ to $x = 4$

C. From $x = 2$ to $x = 3$

D. From $x = 3$ to $x = 4$

1.31 16aeb5f1-86d3-4fcc-b2ef-5ad34f8f79f5 · Easy

The function g is defined by $gx = \sqrt{8x + 1}$. What is the value of $g3$?

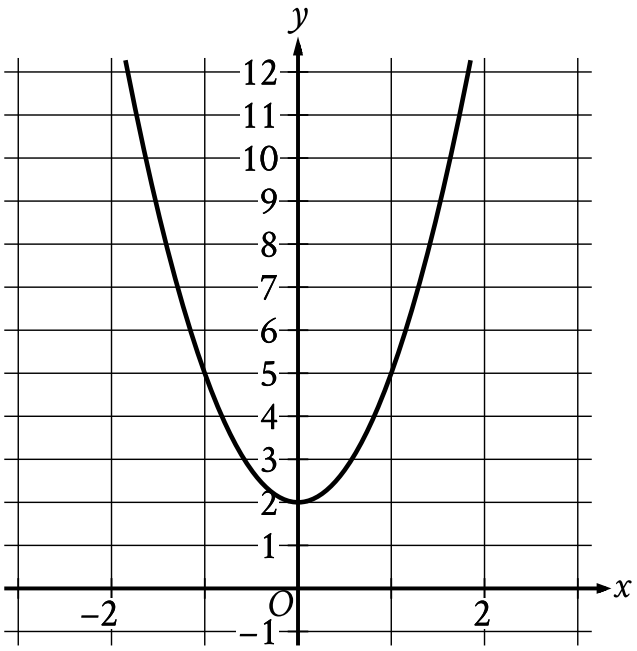
A. $\frac{5}{8}$

B. $\frac{25}{8}$

C. 5

D. 25

1.32 af1af0e8-fb4f-4830-9348-cc9508783406 · Easy



- The parabola opens upward.
- The parabola passes through the following points:
  - (negative 1 comma 5)
  - (0 comma 2)
  - (1 comma 5)

The graph of the quadratic function $y = fx$ is shown. What is the vertex of the graph?

A. $0, -2$

B. $0, -3$

C. $0, 2$

D. $0, 3$

1.33 a31258b0-0750-420d-9cc1-2b4aad1e6dc2 · Easy

The function f is defined by $f(x) = 4 + \sqrt{x}$. What is the value of $f144$?

A. 0

B. 16

C. 40

D. 76

1.34 3c84e10c-eecd-4a7b-b28c-a5de3d455977 · Easy

The function f is defined by $fx = \frac{1}{6x}$. What is the value of fx when $x = 3$?

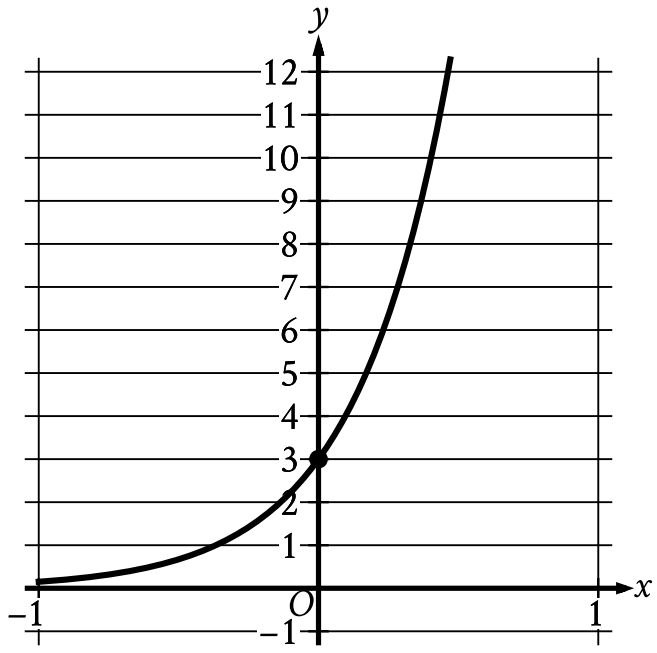
A. $\frac{1}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{9}$

D. $\frac{1}{18}$

1.35 56af9a6d-1c30-4ea5-a528-45b60f366d79 · Easy



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up sharply to the point (0 comma 3).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - approximately (negative 1 comma 0.2)
  - (0 comma 3)

The graph of the exponential function *f* is shown, where $y = fx$. The *y*-intercept of the graph is $0, y$. What is the value of *y*?

Student-produced response: _____

1.36 2c8d6c87-c4db-4555-8418-f31510d6ec98 · Easy

The *y*-intercept of the graph of $y = x^2 + 31$ in the *xy*-plane is $0, y$. What is the value of *y*?

Student-produced response: _____

1.37 05723-DC · Easy

If $f(x)=\frac{x^2-6x+3}{x-1}$, what is $f(-1)$?

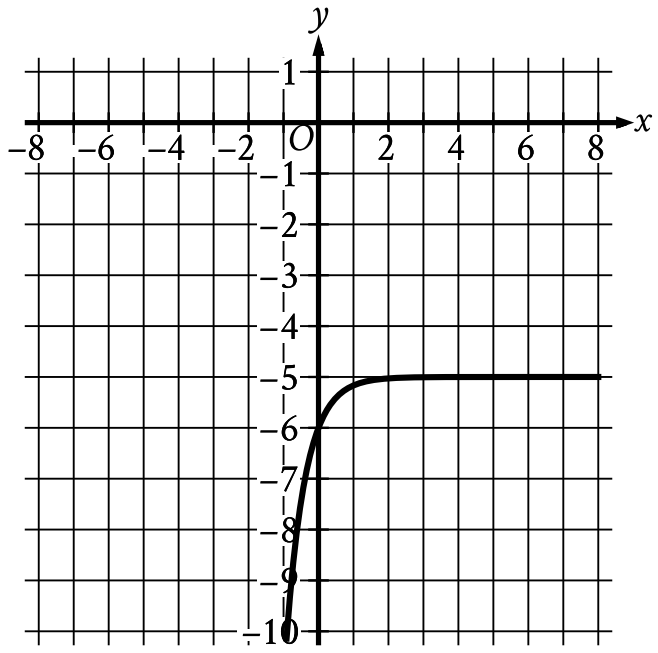
A. –5

B. –2

C. 2

D. 5

1.38 d359151d-6c08-4338-b308-8a73c712373f · Easy



- Moving from left to right:
  - The curve passes from quadrant 3 to quadrant 4.
  - In quadrant 3, the curve trends up sharply to point (0 comma negative 6).
  - In quadrant 4, the curve trends up gradually.
- As x increases, the curve approaches the line y equals negative 5.
- The curve passes through the following points:
  - (0 comma negative 6)
  - (1 comma negative 5 and one sixth)

What is the *y*-intercept of the graph shown?

A. $0, -6$

B. $-6, 0$

C. $0, 0$

D. $-5, -5$

1.39 b0ead772-4a2d-45aa-95e5-204e1ccfdb59 · Easy

The function f is defined by $fx = \frac{16}{x}$. What is the value of fx when $x = 17$?

A. $\frac{16}{17}$

B. $\frac{17}{16}$

C. 16

D. 17

1.40 032723-DC · Easy

$$f(x) = 2(3^x)$$

For the function f defined above, what is the value of $f(2)$?

A. 9

B. 12

C. 18

D. 36

1.41 3412b211-d5f8-400a-988e-f1dd95e0d2e3 · Easy

The function f is defined by $fx = x^3 + 9$. What is the value of $f2$?

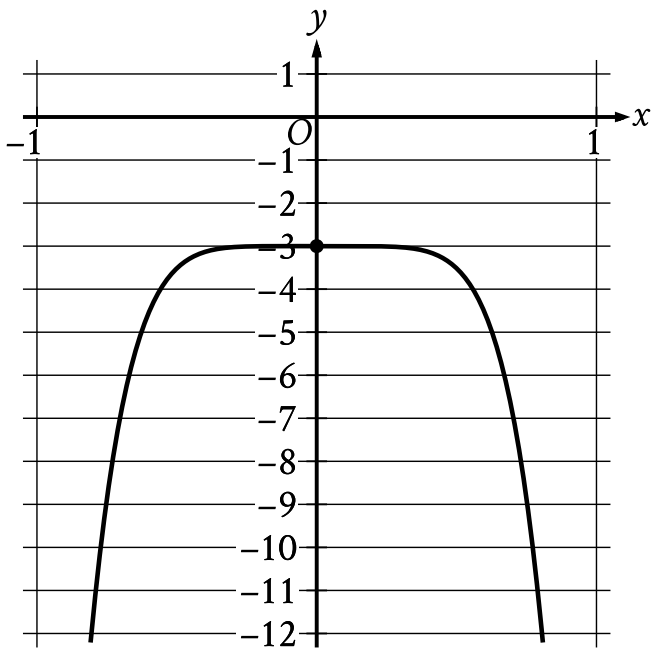
A. 14

B. 15

C. 17

D. 18

1.42 a32cd185-f66c-43e0-b4ca-8c66da898744 · Easy



- In quadrant 3, the curve rises to its maximum at point (0 comma negative 3).
- In quadrant 4, the curve then falls.

The graph of the polynomial function f, where $y = fx$, is shown. The y-intercept of the graph is $0, y$. What is the value of y?

Student-produced response: _____

1.43 e647a4e3-60a1-4bf3-8ceb-4b40a8d430f1 · Easy

The function f is defined by $fx = 7x^3$. In the xy-plane, the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down 2 units. Which equation defines function g?

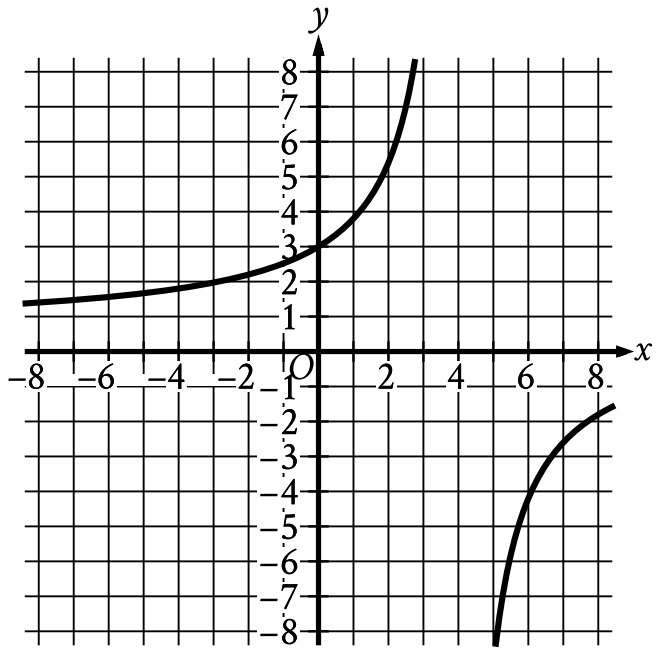
A. $gx = \frac{7}{2}x^3$

B. $gx = 7x^{\frac{3}{2}}$

C. $gx = 7x^3 + 2$

D. $gx = 7x^3 - 2$

1.44 5943941c-677e-4c02-9da5-eddac62bf566 · Easy



- For the first curve:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - In quadrant 2, the curve trends up gradually to point (0 comma 3).
    - In quadrant 1, the curve trends up sharply.
  - As x increases, the curve approaches the line x equals 4.
  - As x decreases, the curve approaches the line y equals 1.
  - The curve passes through the following points:
    - (negative 3 comma 2)
    - (0 comma 3)
- For the second curve:
  - Moving from left to right:
    - The curve is in quadrant 4.
    - In quadrant 4, the curve trends up sharply.
  - As x increases, the curve approaches the line y equals 1.
  - As x decreases, the curve approaches the line x equals 4.
  - The curve passes through the following point:
    - (6 comma negative 4)

The graph of $y = fx$ is shown in the xy-plane. What is the value of $f0$?

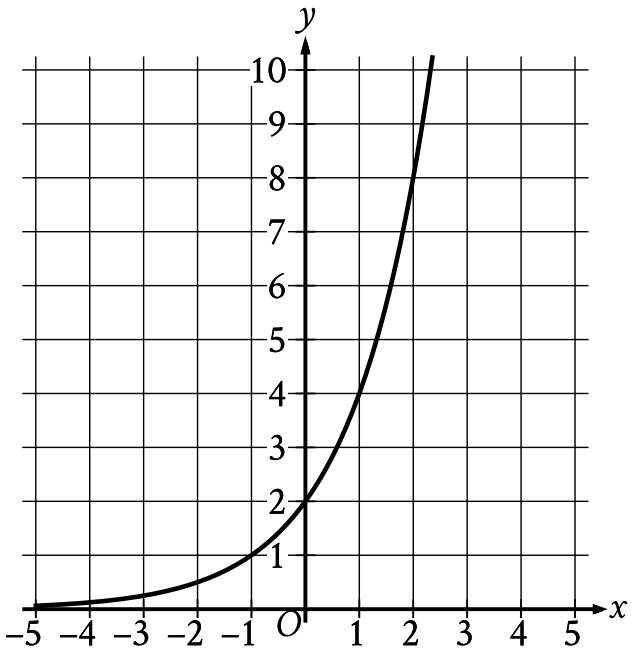
A. -3

B. 0

C. $\frac{3}{5}$

D. 3

1.45 7af6ef83-b534-4900-b63b-a74eee6196f9 · Easy



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 2).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - (negative 1 comma 1)
  - (0 comma 2)
  - (2 comma 8)

What is the y-intercept of the graph shown?

A. $0, 0$

B. $0, 2$

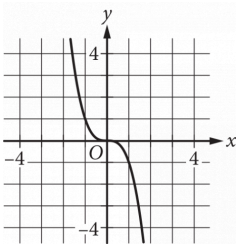
C. $2, 0$

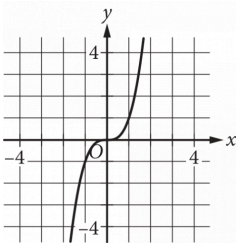
D. $2, 2$

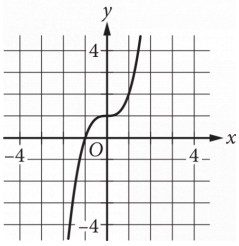
1.46 027806-DC · Easy

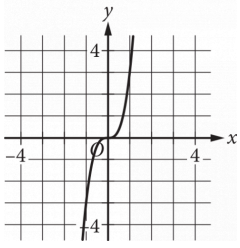
x	y
0	0
1	1
2	8
3	27

The table shown includes some values of x and their corresponding values of y. Which of the following graphs in the xy-plane could represent the relationship between x and y ?

A. 

B. 

C. 

D. 

1.47 08dcb1a6-8ea6-4767-9d5b-83bca79b58fa · Easy

The function f is defined by $fx = x^3 + 15$. What is the value of $f2$?

Student-produced response: _____

1.48 cd3ef442-8790-4484-ae9f-8b639799e10a · Easy

The function g is defined by $g(x) = |x + 18|$. What is the value of $g4$?

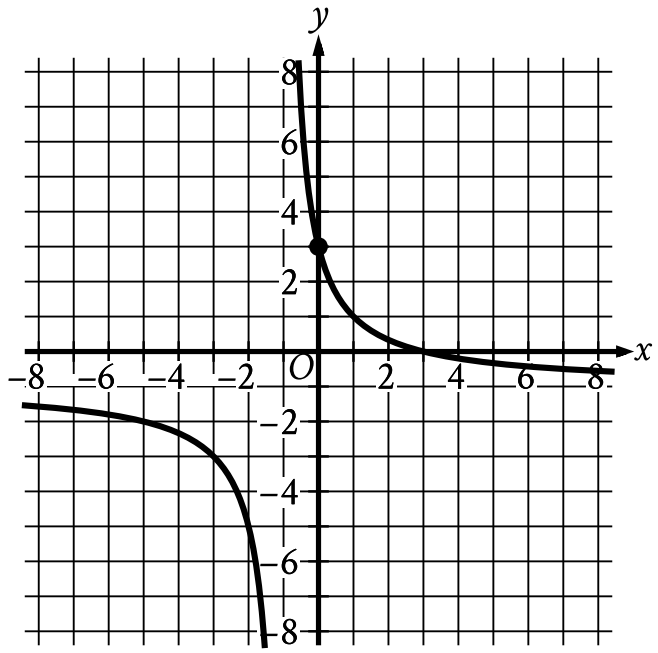
A. -18

B. -4

C. 14

D. 22

1.49 6f8b1a0e-8c22-4208-8038-796a6de480d4 · Easy



- For the first curve:
  - Moving from left to right, the curve is in quadrant 3.
  - As x decreases, the curve approaches the line y equals negative 1.
  - As x increases, the curve approaches the line x equals negative 1.
- For the second curve:
  - Moving from left to right, the curve passes from quadrant 2 to quadrant 1 to quadrant 4.
  - As x decreases, the curve approaches the line x equals negative 1.
  - As x increases, the curve approaches the line y equals negative 1.
- The 2 curves pass through the following points:
  - (negative 6 comma negative nine fifths)
  - (0 comma 3)
  - (4 comma negative one fifth)

What is the *y*-coordinate of the *y*-intercept of the graph shown?

Student-produced response: _____

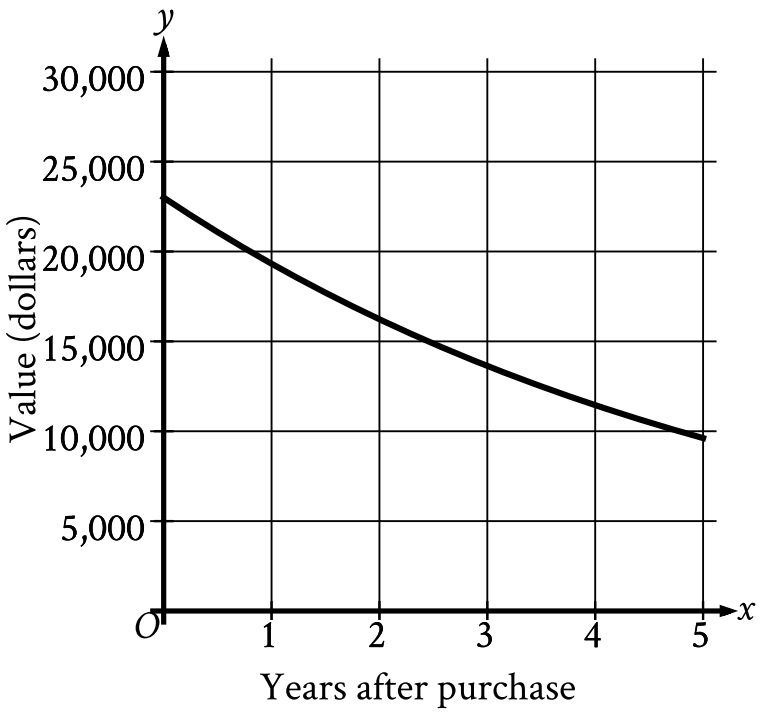
1.50 f7def592-aac8-4d10-bb76-680518e870b2 · Easy

The function f is defined by $f(x) = x^2 + x + 71$. What is the value of $f(2)$?

Student-produced response: _____

1.51 83b9a6a0-309d-476f-bd88-5134eb899234 · Easy

The graph shows the predicted value y, in dollars, of a certain sport utility vehicle x years after it is first purchased.



- The curve is in quadrant 1.
- The curve trends down gradually from left to right.
- The curve passes through the following approximate points:
  - (0 comma 23,000)
  - (1 comma 19,320)
  - (3 comma 13,632)
  - (5 comma 9,619)

Which of the following is closest to the predicted value of the sport utility vehicle 3 years after it is first purchased?

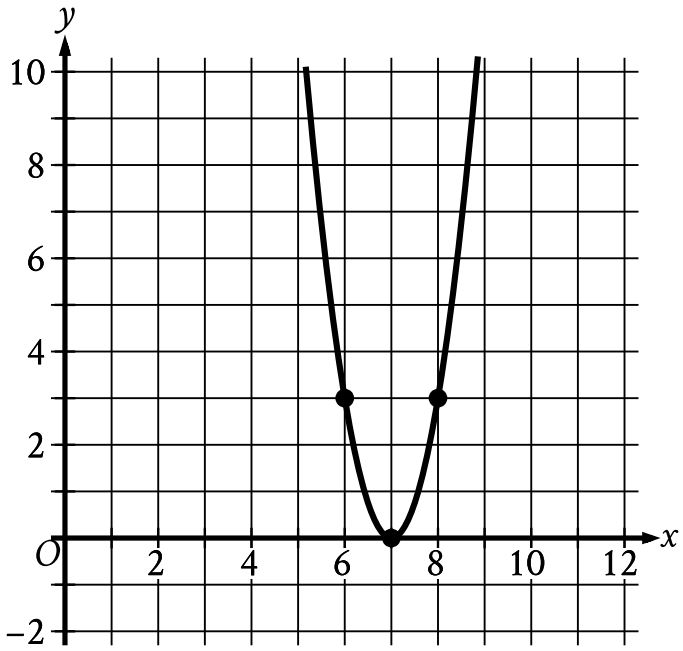
A. $\$ 9{,}619$

B. $\$ 13{,}632$

C. $\$ 19{,}320$

D. $\$ 23{,}000$

1.52 ff1de39f-4c5b-4dcb-8710-09663efcf4f5 · Easy



- The parabola opens upward.
- The vertex is at point (7 comma 0).
- The parabola passes through the following points:
  - (6 comma 3)
  - (7 comma 0)
  - (8 comma 3)

The x-intercept of the graph shown is $x, 0$. What is the value of x?

Student-produced response: _____

1.53 2d504a02-4e7f-409c-a455-32753236fe82 · Easy

At the time of posting a video, a social media channel had 53 subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the number the previous day. Which equation gives the total number of subscribers, n, to the channel d days after the video was posted?

A. $n = 53^d$

B. $n = 53(2)^d$

C. $n = 53\frac{1}{2}^d$

D. $n = 53^2 + d$

1.54 edb987d4-772f-4ea6-8b75-8cd93a8fc5c8 · Easy

A company opens an account with an initial balance of $\$\,36{,}100.00$. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A, where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 13 years is $\$\,68{,}071.93$. Which equation could define A?

A. $At = 36{,}100.001.05^t$

B. $At = 31{,}971.931.05^t$

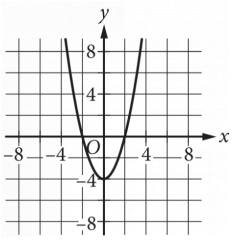
C. $At = 31{,}971.930.05^t$

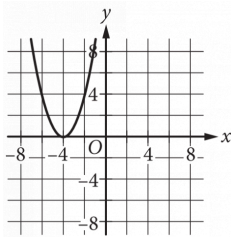
D. $At = 36{,}100.000.05^t$

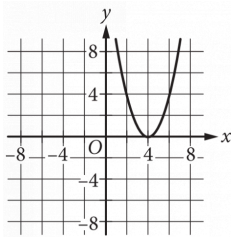
1.55 021441-DC · Easy

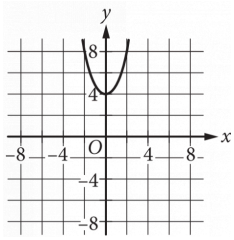
$f(x)=x^2+4$

The function f is defined as shown. Which of the following graphs in the xy-plane could be the graph of $y=f(x)$?

A. 

B. 

C. 

D. 

1.56 5d453a5b-d7a9-40ac-92b4-8a1f6744f99e · Easy

The function $f(x) = 240{,}0001.22^{x}$ gives a company's predicted annual revenue, in dollars, x years after the company started selling jewelry online, where $0 < x \leq 10$. What is the best interpretation of the statement "$f5$ is approximately equal to $648{,}650$" in this context?

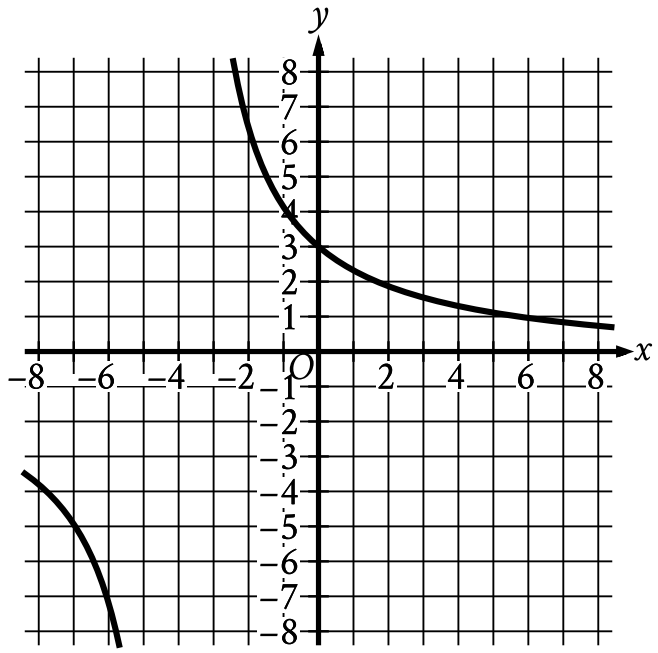
A. 5 years after the company started selling jewelry online, its predicted annual revenue is approximately $648{,}650$ dollars.

B. 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately $648{,}650$ dollars.

C. When the company's predicted annual revenue is approximately $648{,}650$ dollars, it is 5 times the predicted annual revenue for the previous year.

D. When the company's predicted annual revenue is approximately $648{,}650$ dollars, it is 5% greater than the predicted annual revenue for the previous year.

1.57 87aede6d-89e4-4d41-9c3c-af1ae260cc3a · Easy



- For the first curve:
  - Moving from left to right:
    - The curve is in quadrant 3.
    - As x decreases, the curve approaches the line y equals 0.
    - As x increases, the curve approaches the line x equals negative 4.
- For the second curve:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - As x decreases, the curve approaches the line x equals negative 4.
    - As x increases, the curve approaches the line y equals 0.
- The 2 curves pass through the following points:
  - approximately (negative 7 comma negative 4.9)
  - (0 comma 3)
  - approximately (7 comma 0.8)

The graph of $y = fx$ is shown in the xy-plane. The value of $f0$ is an integer. What is the value of $f0$?

Student-produced response: _____

1.58 6343862e-4b55-4888-a763-6c005c8707e0 · Easy

The function f is defined by $fx = 5x^2$. What is the value of $f8$?

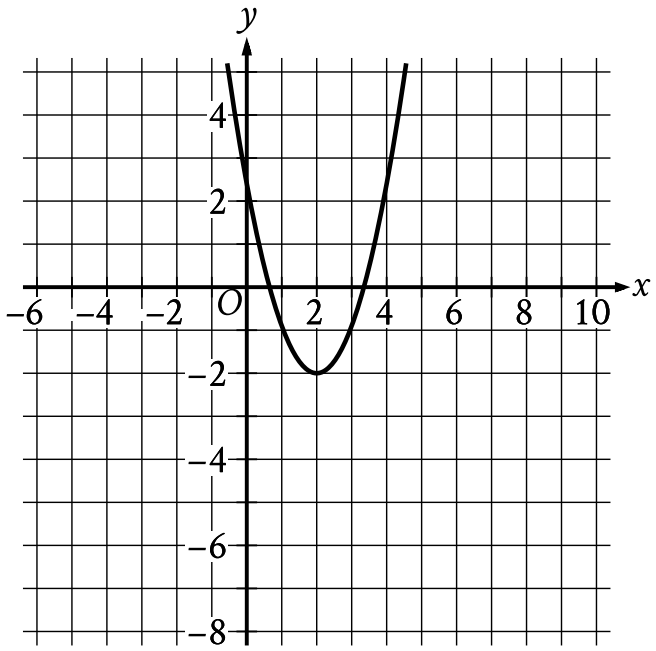
A. 40

B. 50

C. 80

D. 320

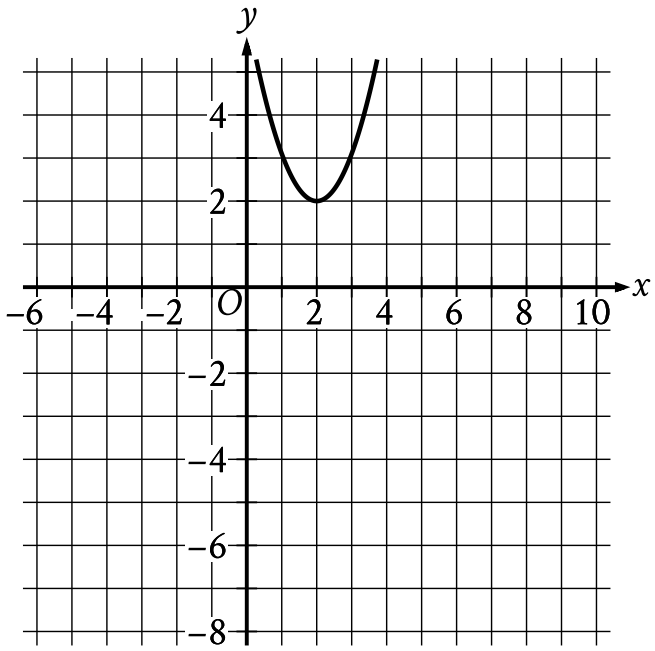
1.59 ec7fe4db-233b-4c49-9a70-2816bda45ba9 · Easy



- The parabola opens upward.
- The vertex is at the point (2 comma negative 2).
- The parabola passes through the following points:
  - (1 comma negative nine tenths)
  - (2 comma negative 2)
  - (3 comma negative nine tenths)

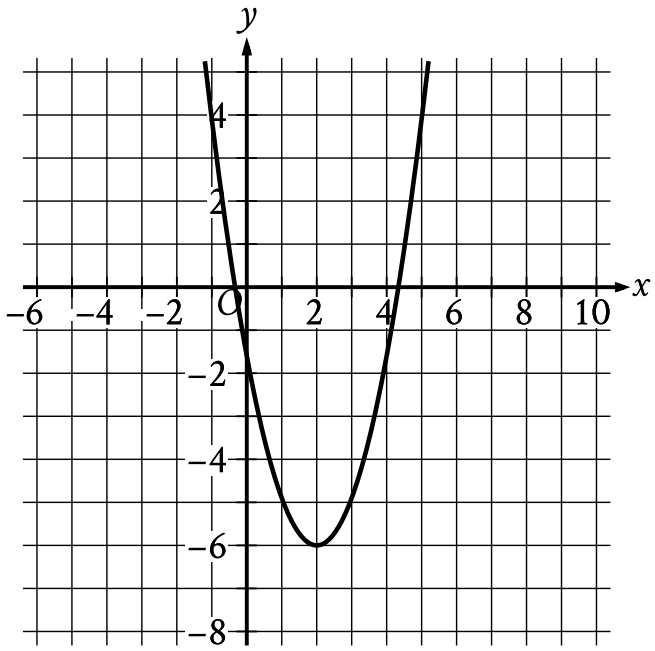
The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

A.



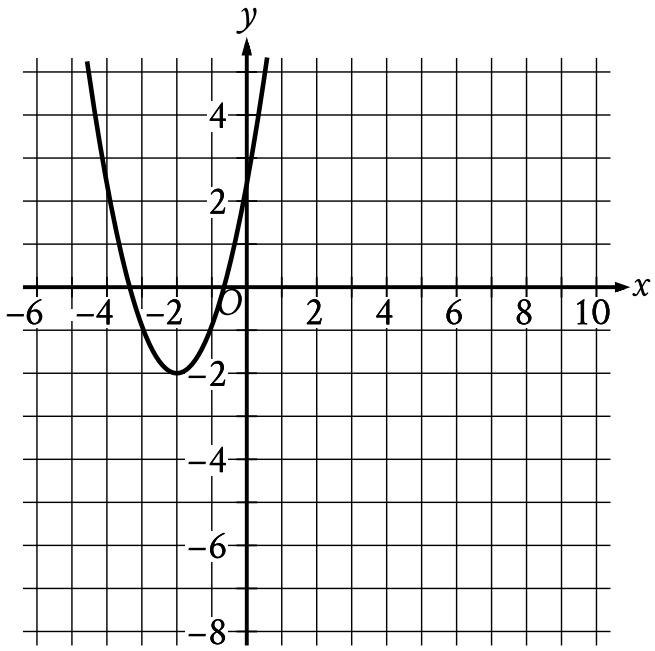
- The parabola opens upward.
- The vertex is at the point (2 comma 2).
- The parabola passes through the following points:
  - (1 comma StartFraction 31 Over 10 EndFraction)
  - (2 comma 2)
  - (3 comma StartFraction 31 Over 10 EndFraction)

B.



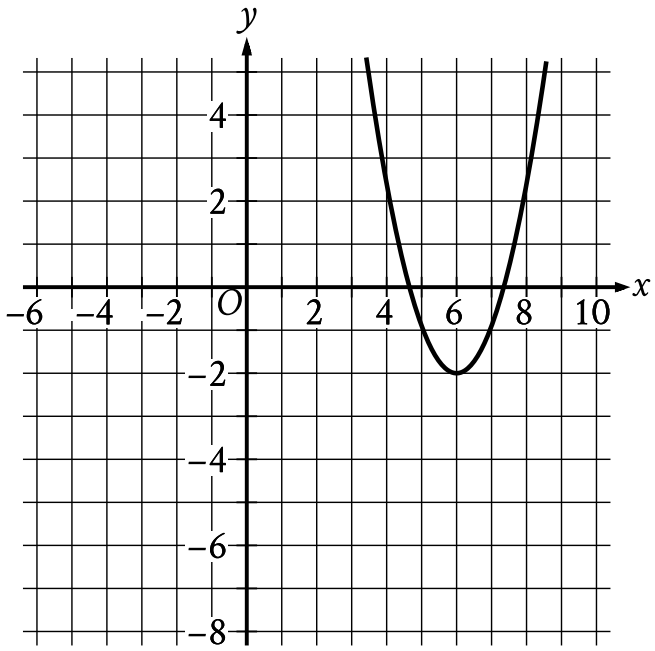
- The parabola opens upward.
- The vertex is at the point (2 comma negative 6).
- The parabola passes through the following points:
  - (1 comma negative StartFraction 49 Over 10 EndFraction)
  - (2 comma negative 6)
  - (3 comma negative StartFraction 49 Over 10 EndFraction)

C.



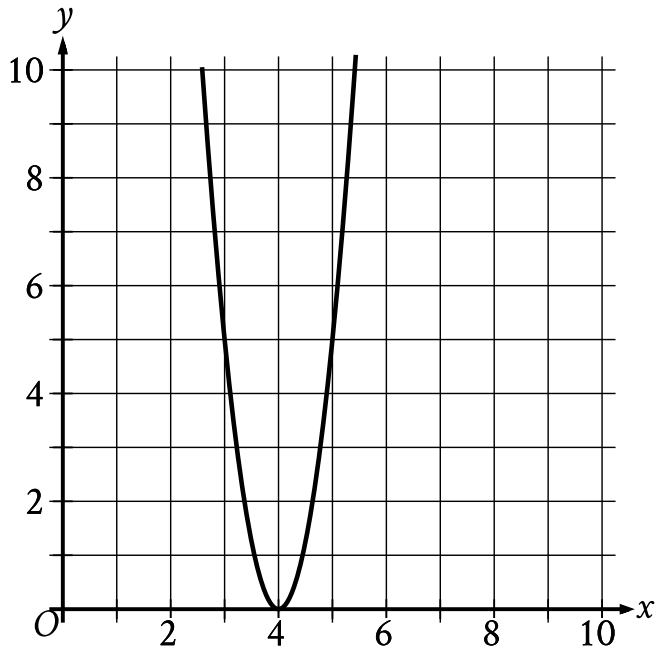
- The parabola opens upward.
- The vertex is at the point (negative 2 comma negative 2).
- The parabola passes through the following points:
    - (negative 3 comma negative nine tenths)
    - (negative 2 comma negative 2)
    - (negative 1 comma negative nine tenths)

D.



- The parabola opens upward.
- The vertex is at the point (6 comma negative 2).
- The parabola passes through the following points:
  - (5 comma negative nine tenths)
  - (6 comma negative 2)
  - (7 comma negative nine tenths)

1.60 52978a8b-cb33-43f8-a130-45f92771e3d0 · Easy



- The parabola opens upward.
- The vertex is at point (4 comma 0).
- The parabola passes through the following points:
  - (3 comma 5)
  - (4 comma 0)
  - (5 comma 5)

What is the x-intercept of the graph shown?

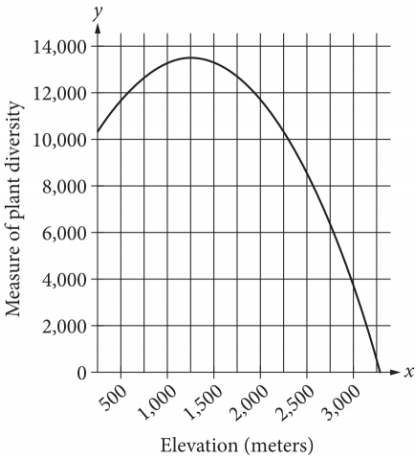
A. $-5, 0$

B. $5, 0$

C. $-4, 0$

D. $4, 0$

1.61 025654-DC · Easy



The quadratic function graphed above models a particular measure of plant diversity as a function of the elevation in a region of Switzerland. According to the model, which of the following is closest to the elevation, in meters, at which plant diversity is greatest?

A. 13,500

B. 3,000

C. 1,250

D. 250

1.62 021520-DC · Easy

$f(x) = (x + 0.25x)(50 - x)$

The function f is defined above. What is the value of $f(20)$?

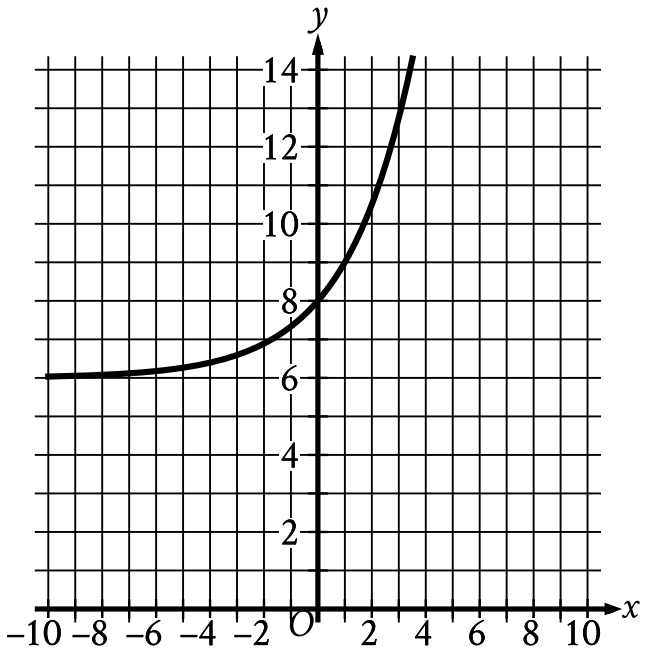
A. 250

B. 500

C. 750

D. 2,000

1.63 2deb518b-5f4a-45ec-828f-20051b7eb7aa · Easy



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 8).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 8)
  - (1 comma 9)

What is the y-intercept of the graph shown?

A. $-8, 0$

B. $-6, 0$

C. $0, 6$

D. $0, 8$

1.64 1cfbd034-f610-4c8e-8e6f-88944150668f · Easy

The function f is defined by $f(x) = 3x^5$. In the xy-plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g?

A. $g(x) = 3x^5 + 4$

B. $g(x) = 3x^5 - 4$

C. $g(x) = 3x^{20}$

D. $g(x) = 12x^5$

1.65 2d05a911-7298-4d94-ad93-ec0fefdedb1d · Easy

There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p, eliminated at the end of round r, where $r \leq 4$?

A. $p = 15\frac{1}{2}^r$

B. $p = 152^r$

C. $p = 240\frac{1}{2}^r$

D. $p = 2402^r$

1.66 ff0ab105-aa85-4cb3-a07f-f00a7c55c4e3 · Medium

The population of the town of Smithville doubled every 75 years from 1659 to 1959. The population of this town was $240{,}000$ in 1959. What was the population of this town in 1659?

Student-produced response: _____

1.67 62b8544c-e4cc-4953-8687-0729c0554f61 · Medium

The function p is defined by $pn = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

A. 2

B. $\frac{8}{3}$

C. 7

D. 8

1.68 00d08c52-5d90-46ea-83d3-f3cc7ab08588 · Medium

The function f is defined by $f(x) = |x - 4x|$. What value of a satisfies $f(5) - f(a) = -15$?

A. -20

B. 5

C. 10

D. 45

1.69 40adf1b6-46fb-41b3-a10c-d6c1ab8c28c8 · Medium

$$y = 3{,}600a^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y, in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

Student-produced response: _____

1.70 b5fc9e88-ea7b-46f4-af78-c536a42d0baf · Medium

$$fx = 4{,}0000.75^{x}$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after 2012, where $x \leq 7$. Which of the following is the best interpretation of $4{,}000$ in this context?

A. The estimated initial beetle population for this species and area in 2012

B. The estimated beetle population for this species and area 7 years after 2012

C. The estimated percent decrease in the beetle population for this species and area each year after 2012

D. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

1.71 62aee8c9-96f8-4f03-aad4-f068bd524749 · Medium

A company has a newsletter. In January 2018, there were $1{,}300$ customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c, subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

A. $c = 1{,}3000.07^{m}$

B. $c = 1{,}3001.07^{m}$

C. $c = 1{,}3001.7^{m}$

D. $c = 1{,}3007^{m}$

1.72 35058beb-41fe-4d27-9257-454cd92cb3f3 · Medium

The function f is defined by $fx = 2700.1^{x}$. What is the value of $f0$?

A. 0

B. 1

C. 27

D. 270

1.73 faf32f9b-26f8-4ad5-8215-b6f52b1118f5 · Medium

For the polynomial function f, the graph of $y = fx$ in the xy-plane passes through the points -50, 10, and 40. Which of the following must be a factor of $f(x)$?

A. $x + 1$

B. $x + 4$

C. $x - 1$

D. $x - 5$

1.74 10571-DC · Medium

$$h(t) = -16t^2 + 110t + 72$$

The function above models the height h, in feet, of an object above ground t seconds after being launched straight up in the air. What does the number 72 represent in the function?

A. The initial height, in feet, of the object

B. The maximum height, in feet, of the object

C. The initial speed, in feet per second, of the object

D. The maximum speed, in feet per second, of the object

1.75 032669-DC · Medium

There were no jackrabbits in Australia before 1788 when 24 jackrabbits were introduced. By 1920 the population of jackrabbits had reached 10 billion. If the population had grown exponentially, this would correspond to a 16.2% increase, on average, in the population each year. Which of the following functions best models the population $p(t)$ of jackrabbits t years after 1788?

A. $p(t) = 1.162(24)^t$

B. $p(t) = 24(2)^{1.162t}$

C. $p(t) = 24(1.162)^t$

D. $p(t) = (24 \cdot 1.162)^t$

1.76 91bb6f2f-9d21-4a10-add7-5e0ad39aece8 · Medium

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

A.

x	y
-6	-9
-5	-8
4	1

B.

x	y
-6	-3
-5	-3
4	-3

C.

x	y
-6	-3
-5	-2
4	7

D.

x	y
-6	3
-5	3
4	3

1.77 bcebeee7-1522-4888-bc6b-06a4c5b1dae3 · Medium

A function p estimates that there were $2{,}000$ animals in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of animals in this population increased by 3% of the number of animals in the population the previous year. Which equation defines this function, where px is the estimated number of animals in the population x years after 1998?

A. $px = 2{,}0003^{x}$

B. $px = 2{,}0001.97^{x}$

C. $px = 2{,}0001.03^{x}$

D. $px = 2{,}0000.97^{x}$

1.78 d1baa89a-a42c-46e1-a485-7574f59a8753 · Medium

The number of bacteria in a liquid medium doubles every day. There are $44{,}000$ bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria, y, in the liquid medium t days after the start of the observation?

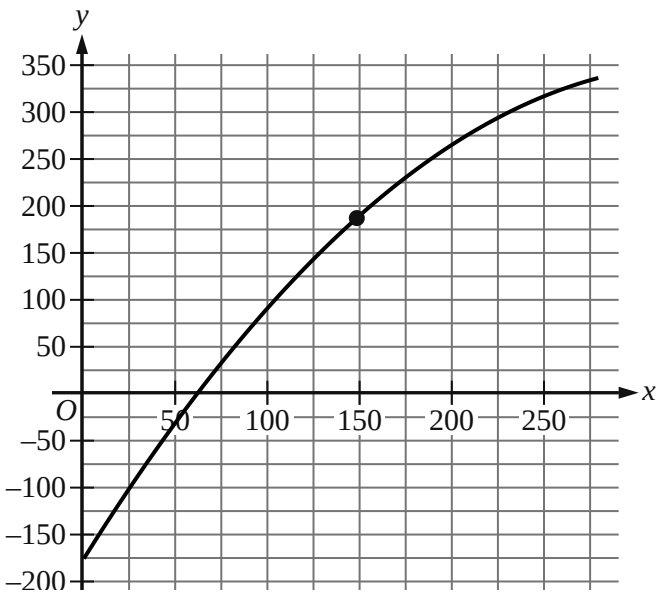
A. $y = \frac{1}{2}44{,}000^{t}$

B. $y = 244{,}000^{t}$

C. $y = 44{,}000\frac{1}{2}^{t}$

D. $y = 44{,}0002^{t}$

1.79 c4bea442-d7d0-404b-a78a-6881543f941b · Medium



- The parabola opens downward.
- The parabola passes through the following approximate points:
  - (50 comma negative 33.83)
  - (149.02 comma 186.05)
  - (250 comma 315.37)

The graph shows the estimated boiling point y, in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $149.02186.05$?

A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.

B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

1.80 af32328c-4e46-4a53-b9c0-f627a4d608f9 · Medium

Bacteria are growing in a liquid growth medium. There were $300{,}000$ cells per milliliter during an initial observation. The number of cells per milliliter doubles every 3 hours. How many cells per milliliter will there be 15 hours after the initial observation?

A. $1{,}500{,}000$

B. $2{,}400{,}000$

C. $4{,}500{,}000$

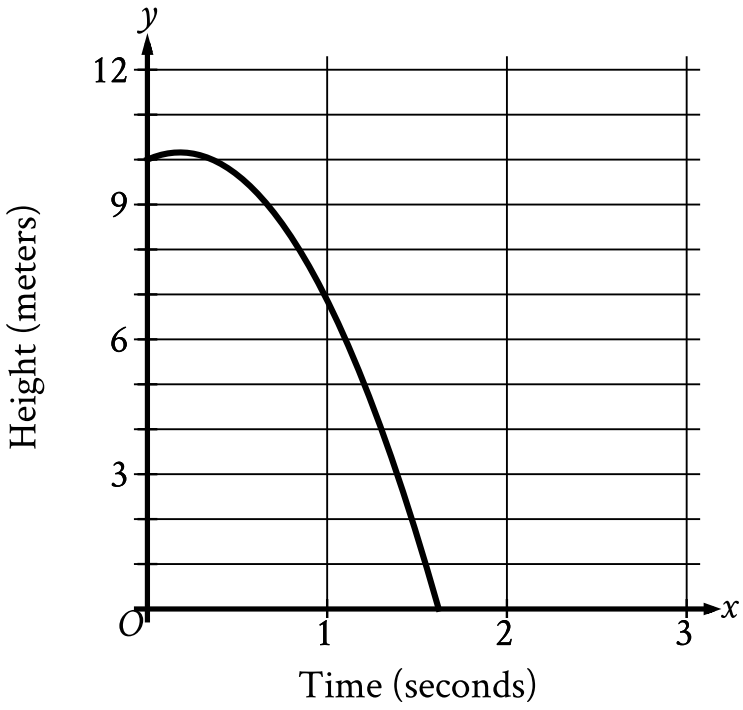
D. $9{,}600{,}000$

1.81 783877b4-b840-4260-8d01-4234651d93c4 · Medium

What is an x-coordinate of an x-intercept of the graph of $y = 3x - 14x + 5x + 4$ in the xy-plane?

Student-produced response: _____

1.82 2b50d060-8469-4554-937e-da992751dc98 · Medium



- The parabola opens downward.
- The vertex is at the approximate point (0.2 comma 10.2).
- The parabola passes through the following points:
  - (0 comma 10)
  - approximately (0.2 comma 10.2)
  - approximately (1 comma 6.9)
  - approximately (1.6 comma 0)

A competitive diver dives from a platform into the water. The graph shown gives the height above the water y, in meters, of the diver x seconds after diving from the platform. What is the best interpretation of the x-intercept of the graph?

A. The diver reaches a maximum height above the water at 1.6 seconds.

B. The diver hits the water at 1.6 seconds.

C. The diver reaches a maximum height above the water at 0.2 seconds.

D. The diver hits the water at 0.2 seconds.

1.83 076cf150-d233-4c88-b49e-49a87889ba92 · Medium

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s(87) = 23$ in this context?

A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.

B. The mass of the sample has decreased by approximately 23 grams 87 years after the sample starts to decay.

C. The mass of the sample has decreased by approximately 87 grams 23 years after the sample starts to decay.

D. Approximately 87 grams of the sample remains 23 years after the sample starts to decay.

1.84 e3e1bd01-4a4e-4c9c-8e91-28a5165667b6 · Medium

$$f(x) = x + 6x - 4$$

If the given function f is graphed in the xy-plane, where $y = f(x)$, what is the x-coordinate of an x-intercept of the graph?

Student-produced response: _____

1.85 e78c85a7-41e7-485b-b784-99727df47183 · Medium

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Rosa opened the account and the total amount n, in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n?

A. $n = 1 + 604^t$

B. $n = 1 + 0.004^t$

C. $n = 6041 + 0.004^t$

D. $n = 0.0041 + 604^t$

1.86 b790c80c-3eb0-4ef6-8101-d653f1487b32 · Medium

$$mt = -0.0274\frac{t}{7}^2 + 7.3873\frac{t}{7} + 75.032$$

The function m gives the predicted body mass mt, in $\text{kilograms}\quad \text{kg}$, of a certain animal t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement "$m330$ is approximately equal to 362" in this context?

A. The predicted body mass of the animal was approximately $330\quad \text{kg}$ 362 days after it was born.

B. The predicted body mass of the animal was approximately $362\quad \text{kg}$ 330 days after it was born.

C. The predicted body mass of the animal was approximately $362\quad \text{kg}$ $\frac{330}{7}$ days after it was born.

D. The predicted body mass of the animal was approximately $\frac{330}{7}\quad \text{kg}$ 362 days after it was born.
